

Canalicular method rebuilt around the physical scale

=====

PIXEL SIZE 0.13 um/px. PIXEL SIZE 0.13 UM/PX, ROUNDED, UNCONFIRMED.

PRE-VALIDATION. No variant is called better than another.

Variants:

A current method, as is
P1 ridge, sigmas 0.13, 0.17, 0.21 um
P2 ridge, sigmas 0.17, 0.21, 0.26 um (primary)
P3 ridge, one scale per image chosen from the image
P4 P2 with the low cut at 0.9 of the high cut
P5 P2 with gap bridging off
N P2 on the notch filtered image

Mean over the 8 images

measure	A	P1	P2	P3	P4	P5	N
width_fwhm_median_um	0.6368	0.6314	0.6352	0.6409	0.6362	0.6351	0.6352
width_dt_median_um	0.7853	0.5353	0.6006	0.7465	0.5814	0.6006	0.6006
low_contrast_fraction	0.02063	0.05829	0.04053	0.03504	0.02528	0.04019	0.04044
contrast_snr_median	9.955	7.299	7.877	8.756	7.724	7.882	7.877
straight_run_fraction	0.221	0.2267	0.2375	0.247	0.2318	0.2365	0.2376
straight_run_fraction_smoothed	0.1872	0.1797	0.194	0.2038	0.1921	0.1931	0.1939
rectangle_count	0	0.25	0	0	0	0	0
connected_to_lacuna_fraction	0.336	0.4655	0.48	0.5373	0.4341	0.4557	0.48
junction_count	529.8	791.2	738.8	705.9	608.6	733.6	738.1
thread_end_fraction	0.7641	0.6959	0.6869	0.6671	0.716	0.6927	0.6871
field_length_density_um_per_um2	0.3008	0.3358	0.3237	0.3106	0.3029	0.3232	0.3236
roots_per_cell	7.545	8.279	7.965	7.508	7.483	7.965	7.965
skeleton_px	3.992e+04	4.456e+04	4.295e+04	4.122e+04	4.019e+04	4.289e+04	4.294e+04
n_bridges	18.12	21.25	19.62	21	85.12	0	19.38
legacy_unsupported_fraction	0.06798	0.1018	0.09214	0.08723	0.06008	0.09118	0.09202

Change from A, per cent

measure	A	P1	P2	P3	P4	P5	N
width_fwhm_median_um	+0.0	-0.9	-0.3	+0.6	-0.1	-0.3	-0.3
width_dt_median_um	+0.0	-31.8	-23.5	-4.9	-26.0	-23.5	-23.5
low_contrast_fraction	+0.0	+182.6	+96.5	+69.9	+22.5	+94.8	+96.1
contrast_snr_median	+0.0	-26.7	-20.9	-12.0	-22.4	-20.8	-20.9
straight_run_fraction	+0.0	+2.6	+7.5	+11.8	+4.9	+7.0	+7.5
straight_run_fraction_smoothed	+0.0	-4.0	+3.6	+8.9	+2.6	+3.2	+3.6
rectangle_count							
connected_to_lacuna_fraction	+0.0	+38.5	+42.9	+59.9	+29.2	+35.6	+42.8
junction_count	+0.0	+49.4	+39.5	+33.2	+14.9	+38.5	+39.3
thread_end_fraction	+0.0	-8.9	-10.1	-12.7	-6.3	-9.4	-10.1
field_length_density_um_per_um2	+0.0	+11.6	+7.6	+3.3	+0.7	+7.5	+7.6
roots_per_cell	+0.0	+9.7	+5.6	-0.5	-0.8	+5.6	+5.6
skeleton_px	+0.0	+11.6	+7.6	+3.3	+0.7	+7.5	+7.6
n_bridges	+0.0	+17.2	+8.3	+15.9	+369.7	-100.0	+6.9
legacy_unsupported_fraction	+0.0	+49.7	+35.5	+28.3	-11.6	+34.1	+35.4

pixel size 0.13 um/px, rounded, unconfirmed.

Pre-validation: no number here has been checked against a manual count.

A diagnosis and a better founded experiment. No variant is called better than another.

width_fwhm_*: the full width at half maximum of the intensity profile across the thread, taken on the flattened image along the normal at each usable skeleton pixel. It INCLUDES the optical

blur, so it is an apparent width in the image and an upper bound on the diameter, not the diameter. No point spread function is computed or subtracted: the numerical aperture, the emission wavelength and the pinhole are unknown.

width_dt_*: the older measure, twice the distance transform of the mask, kept beside it. It can only return the width of the mask it is given.

contrast_snr: the intensity at a skeleton pixel minus the median intensity 4 to 6 px away along the normal on both sides, over the noise, where the noise is 1.4826 times the median absolute deviation of the flattened image outside the lacunae, the vascular regions and that variant's own mask. low_contrast_fraction is the share below 2.

legacy_unsupported_fraction: the earlier measure, which used the CURRENT method's own threshold and so favoured the current method. Kept only so the earlier tables can be followed.

straight_run_fraction: share of skeleton pixels in an axis-aligned run of at least 8 px. A thin skeleton on a pixel grid contains straight runs by its nature, and the current method already scores about 0.22, so only differences between variants mean anything here.

straight_run_fraction_smoothed: the same after each branch path is smoothed with a 5 point moving average, which removes most of the pixel grid effect.